

A sharp answer to the large-block spectral-gap question in arXiv:1607.05674

Candidate full solution packet

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Status. `candidate_full_solution_likely_valid`. The proof identifies an inequality-direction inconsistency in the source and then sharpens its estimates to determine the exact off-target Fourier coefficient.

1 Question and answer

Embed $U(k)$ in $U(n)$ as $a \mapsto a \oplus I_{n-k}$, let $m_{U(k)}$ be its Haar probability, and define Pisier's central probability

$$\mu_{k,n} = \int_{U(n)} \delta_s * m_{U(k)} * \delta_{s^{-1}} dm_{U(n)}(s).$$

Let σ_n be the defining representation and $S_n = \{\sigma_n, \overline{\sigma_n}\}$. If $\ell = n - k$, then

$$\widehat{\mu}_{k,n}(\sigma_n) = \widehat{\mu}_{k,n}(\overline{\sigma_n}) = \frac{\ell}{n} I.$$

Remark 2.11 of arXiv:1607.05674v7 asks whether a comparable spectral gap holds when $k = [\theta n]$ and $1/2 < \theta < 1$.

Proof of Theorem 2.1 We apply first part (i) in Lemma 2.8 to settle the case $d = 0$. Thus we may assume $d \geq 1$. We apply then part (ii) from that same Lemma 2.8 to settle the cases either $n = 2k$ or $n = 2k - 1$, with the restriction $n - 1 > k$ which requires $n > 3$. Then Lemma 2.10 settles the remaining case $n = 2k + 1$. Note that $k/n \rightarrow 1/2$ when $n \rightarrow \infty$ if either $k = n/2$, $k = k_+$ or $k = k_-$, and all the bounds appearing in Lemmas 2.8 and 2.10 tend to $1/4$. Therefore, for any $1/4 < \gamma < 1/2$ there is $n(\gamma)$ such that for any $n \geq n(\gamma)$

$$\sup_{\pi \notin S_n} \|\widehat{\nu}_n(\pi)\| \leq \gamma.$$

Since $\widehat{\mu}_{k,n}(\pi) = k/n$ when $\pi = \sigma_n$ or $\pi = \overline{\sigma_n}$ (and since $(k_+ + k_-)/2n = 1/2$) we have $\widehat{\nu}_n(\pi) = 1/2$. Thus ν_n has a $(1/2, \gamma)$ -spectral gap for any $n \geq n(\gamma)$, which settles the last assertion in Theorem 2.1. Checking the bounds for small values of n , actually we can find a $\gamma < 1/2$ whenever $n \geq 4$. $\square$

Remark 2.11 (A natural question). Assume that $k = [\theta n]$ where $0 < \theta < 1$ is fixed. Then $\widehat{\mu}_{k,n}(\sigma_n) = \widehat{\mu}_{k,n}(\overline{\sigma_n}) = (n - k)/n \approx 1 - \theta$. By Lemmas 2.8 and 2.10, if we assume $\theta \leq 1/2$ (to ensure that $k \leq n - k$) then $\mu_{k,n}$ has a (δ_n, γ_n) -spectral gap with $\delta_n \approx 1 - \theta$ and $\gamma_n \approx (1 - \theta)^2$ when $n \rightarrow \infty$. We do not know whether this (or any similar spectral gap) holds when $1/2 < \theta < 1$.

Theorem 1 (sharp large-block gap). *Let $1 \leq \ell = n - k \leq k$. Then $\mu_{k,n}$ has a $(\delta_{k,n}, \gamma_{k,n})$ -spectral gap with respect to S_n , where*

$$\delta_{k,n} = \frac{\ell}{n}, \quad \gamma_{k,n} = \frac{\ell(\ell+1)}{n(n+1)}.$$

Moreover, the second constant is exact:

$$\sup_{\substack{\pi \in \widehat{U(n)}, \pi \neq 1 \\ \pi \notin S_n}} \|\widehat{\mu}_{k,n}(\pi)\| = \frac{\ell(\ell+1)}{n(n+1)}.$$

The maximum is attained by $\text{Sym}^2(\sigma_n)$ and its conjugate.

Since $(\ell+1)/(n+1) < 1$, one has $\gamma_{k,n} < \delta_{k,n}$. For $k = \lfloor \theta n \rfloor$ with $1/2 < \theta < 1$,

$$\delta_{k,n} \longrightarrow 1 - \theta, \quad \gamma_{k,n} \longrightarrow (1 - \theta)^2.$$

Thus the answer is affirmative with exactly the square asymptotics requested in the source.

2 The source reduction

We use the notation and formulas of Sections 2.6–2.8 of the source. An irreducible representation is indexed by $m = (m_1 \geq \dots \geq m_n) \in \mathbb{Z}^n$. If $m_n \leq 0$, put

$$d = -m_n, \quad \lambda_i = m_i + d,$$

so λ is a partition with $\lambda_n = 0$, and let $[d]^k = (d, \dots, d, 0, \dots, 0)$, with d repeated k times. The source's Lemma 2.5 gives

$$\widehat{\mu}_{k,n}(\pi_m) = \frac{s_{\lambda/[d]^k}(1^\ell)}{s_\lambda(1^n)} I \tag{1}$$

when $[d]^k \subset \lambda$, and zero otherwise. It is also zero if $m_n > 0$. Hence all coefficients in question are nonnegative scalars.

The decisive literal observation is that part (ii) of the source's Lemma 2.8 assumes

$$n - k \leq k,$$

exactly the range later called unknown. Remark 2.11 instead says the lemmas apply when $k \leq n - k$. Even the source's four displayed bounds in part (ii) are strictly below ℓ/n , so they already yield some spectral gap in the purportedly unknown range. We now make the strongest of those estimates sharp.

Put

$$A = \frac{\ell(\ell+1)}{n(n+1)}, \quad B = \frac{\ell(\ell-1)}{n(n-1)}.$$

The cases in Lemma 2.8 give the following.

- If $d = 0$, then every off-target nonzero coefficient is at most A when $\lambda_1 \geq 2$, and at most B when $\lambda_1 = 1$.
- If $d \geq 1$ and $\lambda_k \geq 2$, the coefficient is at most A .
- If $d = 1$, $\lambda_k = 1$, and $\lambda_1 = 1$ but $\lambda \neq (1^{n-1}, 0)$, it is at most B .
- If $d = 1$, $\lambda_k = 1$, $\lambda_1 \geq 2$, and $\lambda_{n-1} = 0$, it is at most B . The source assumes $k < n - 1$ in this line; when $k = n - 1$ the case is impossible because $\lambda_k = \lambda_{n-1} = 1$.

Since $B \leq A$, only the remaining mixed case needs attention.

3 The sharp mixed-case estimate

Lemma 1. *Assume $\ell \leq k$, $d = 1$, $\lambda_k = 1$, $\lambda_1 \geq 2$, and $\lambda_{n-1} \geq 1$. Then*

$$\frac{s_{\lambda/[1]^k}(1^\ell)}{s_\lambda(1^n)} \leq \frac{\ell^2}{n^2 - 1} \leq A.$$

Proof. Monotonicity of λ and $\lambda_k = 1$ show that $\lambda_k = \dots = \lambda_{n-1} = 1$. For $i < n$, put $\alpha_i = \lambda_i - 1$. In the skew diagram $\lambda/[1]^k$, row k is empty. Above it lies the diagram α , shifted one column to the right; below it lies a column of length $\ell - 1$. The components are disjoint, so the skew Schur function factors:

$$s_{\lambda/[1]^k}(1^\ell) = s_\alpha(1^\ell) s_{(1^{\ell-1})}(1^\ell) = \ell s_\alpha(1^\ell). \quad (2)$$

If $s_\alpha(1^\ell) = 0$, the conclusion is immediate. Otherwise α has at most ℓ nonzero rows; pad it to $\alpha = (\alpha_1, \dots, \alpha_\ell)$.

Weyl's dimension product, with the factors inside the first ℓ rows cancelled against $s_\alpha(1^\ell)$, gives

$$\frac{s_\lambda(1^n)}{s_\alpha(1^\ell)} = k \prod_{i=1}^{\ell} \left[\prod_{j=\ell+1}^{n-1} \frac{\alpha_i + j - i}{j - i} \right] \frac{\alpha_i + n - i + 1}{n - i}. \quad (3)$$

Every factor on the right is nondecreasing in its corresponding α_i . Since $\lambda_1 \geq 2$, the partition α is nonzero, so the product is minimized at $\alpha = (1, 0, \dots, 0)$. At that point the three telescoping products give

$$\frac{s_\lambda(1^n)}{s_\alpha(1^\ell)} \geq k \cdot \frac{n-1}{\ell} \cdot \frac{n+1}{n-1} \cdot \frac{n-1}{k} = \frac{n^2 - 1}{\ell}.$$

Combining this with (2) proves the first inequality. Finally,

$$\frac{\ell^2}{n^2 - 1} \leq \frac{\ell(\ell + 1)}{n(n + 1)}$$

is equivalent to $\ell \leq n - 1$. □

4 Proof of the sharp theorem

Proof of Theorem 1. Formula (1), the zero cases surrounding it, and the exhaustive Lemma 2.8 split listed above cover every irreducible representation. The trivial representation is omitted; the partitions $(1, 0, \dots, 0)$ with $d = 0$ and $(1, \dots, 1, 0)$ with $d = 1$ are precisely σ_n and $\overline{\sigma}_n$ and have coefficient ℓ/n . Every other coefficient is bounded by A , by the source estimates and Lemma 1.

It remains to prove sharpness. For $\pi = \text{Sym}^2(\sigma_n)$, the $U(k)$ -fixed subspace is $\text{Sym}^2(\mathbb{C}^\ell)$, because $\mathbb{C}^n = \mathbb{C}^k \oplus \mathbb{C}^\ell$ and $U(k)$ acts trivially on the second summand. Centralization turns the fixed-space projection into the scalar normalized rank, so

$$\widehat{\mu}_{k,n}(\pi) = \frac{\dim \text{Sym}^2(\mathbb{C}^\ell)}{\dim \text{Sym}^2(\mathbb{C}^n)} I = \frac{\binom{\ell+1}{2}}{\binom{n+1}{2}} I = AI.$$

The conjugate representation gives the same value. This proves equality in the supremum and completes the theorem. □

5 Verification and novelty boundary

An independent exact-arithmetic script evaluates the numerator in (1) by the Jacobi–Trudi determinant and the denominator by Weyl’s formula. Bounded partition enumeration for $2 \leq n \leq 9$ and all $\lceil n/2 \rceil \leq k < n$ found the exact maximum A in every case. This check is supplementary; no finite enumeration is used in the proof.

The bounded novelty audit searched the arXiv id, exact title, source remark, measure notation, sharp formula, and later unitary spectral-gap literature. The four works indexed as citing the published chapter use it as Sidon/Paley background and do not discuss Remark 2.11. The closest later primary source is G. Alon and D. Puder, *Aldous-type Spectral Gaps in Unitary Groups*, arXiv:2603.00353. It studies noncentral sums of projections obtained from coordinate-block Haar measures. Its mean-field theorem determines the smallest eigenvalue of that noncentral Laplacian, whereas the result here determines the normalized rank of one block-fixed subspace after averaging over all unitary conjugates. That paper does not cite the source or state the formula above. No erratum or later exact answer was located through 2026-08-11.

The conservative mathematical claim is therefore a candidate full sharp answer pending human review. Even if the inequality reversal in Remark 2.11 is regarded as merely clerical, the exact off-target maximum and its disconnected-skew proof are a material strengthening not stated in the source.

References

1. G. Pisier, *Spectral gap properties of the unitary groups: around Rider’s results on non-commutative Sidon sets*, arXiv:1607.05674v7; *Operator Theory: Advances and Applications* 261 (2018), 417–465, doi:10.1007/978-3-319-59078-3_21.
2. G. Alon and D. Puder, *Aldous-type Spectral Gaps in Unitary Groups*, arXiv:2603.00353v1 (2026).